Group8_ANN_Lab3

April 30, 2026

1 Group 8 - ANN Lab 3 (Pre-trained CNNs)

This notebook answers all required items in one file: 1. Hyperparameter tuning of `ImageDataGenerator`. 2. VGG19 implementation and comparison with VGG16. 3. VGG16 test accuracy vs training sample sizes `[500, 1000, 2000, 5000, 10000, 15000]` (no augmentation, 30 epochs/run). 4. Xception (Extreme Inception) implementation and comparison with VGG models.

Each training item saves model artifacts as `.h5`.

[51]:
```
import os
import random
import shutil
import time
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf

from IPython.display import Markdown, display
from tensorflow.keras import layers, models, optimizers
from tensorflow.keras.callbacks import EarlyStopping
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing import image as keras_image
from tensorflow.keras.applications import VGG16, VGG19, Xception

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)

PROJECT_DIR = Path.cwd()
DATA_DIR = PROJECT_DIR / 'data'
OUTPUT_DIR = PROJECT_DIR / 'models'
OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
```

```python
BASE_DATASET_DIR = DATA_DIR / 'cats_and_dogs_from_petimages'
WORKING_SPLIT_DIR = DATA_DIR / 'working_split_for_lab3'

print('TensorFlow version:', tf.__version__)
print('Project directory:', PROJECT_DIR)
print('Base dataset:', BASE_DATASET_DIR)
print('Output directory:', OUTPUT_DIR)
```

```
TensorFlow version: 2.21.0
Project directory: /home/solyvie/workspace/cs-
projects/25263CSDMachineLearning/[7] Group 8 - Module 6 CNN Part 3 - Using Pre-
Trained Networks
Base dataset: /home/solyvie/workspace/cs-projects/25263CSDMachineLearning/[7]
Group 8 - Module 6 CNN Part 3 - Using Pre-Trained
Networks/data/cats_and_dogs_from_petimages
Output directory: /home/solyvie/workspace/cs-
projects/25263CSDMachineLearning/[7] Group 8 - Module 6 CNN Part 3 - Using Pre-
Trained Networks/models
```

[52]:
```python
def count_split(split_dir: Path):
    counts = {}
    total = 0
    for cls in ['cats', 'dogs']:
        cls_dir = split_dir / cls
        n = len(list(cls_dir.glob('*'))) if cls_dir.exists() else 0
        counts[cls] = n
        total += n
    counts['total'] = total
    return counts


def print_split_table(title, split_map):
    rows = []
    for split_name, split_dir in split_map.items():
        c = count_split(split_dir)
        rows.append({
            'split': split_name,
            'cats': c['cats'],
            'dogs': c['dogs'],
            'total': c['total'],
            'path': str(split_dir)
        })
    df = pd.DataFrame(rows)
    print(f'\n{title}')
    display(df)
    return df
```

```python
source_train = BASE_DATASET_DIR / 'train'
source_val = BASE_DATASET_DIR / 'validation'
source_test = BASE_DATASET_DIR / 'test'

if source_test.exists():
    train_dir = source_train
    validation_dir = source_val
    test_dir = source_test
    split_note = 'Using dataset-provided train/validation/test directories.'
else:
    if WORKING_SPLIT_DIR.exists():
        shutil.rmtree(WORKING_SPLIT_DIR)

    for split in ['train', 'validation', 'test']:
        for cls in ['cats', 'dogs']:
            (WORKING_SPLIT_DIR / split / cls).mkdir(parents=True, exist_ok=True)

    for cls in ['cats', 'dogs']:
        src_cls = source_train / cls
        dst_cls = WORKING_SPLIT_DIR / 'train' / cls
        for f in src_cls.iterdir():
            if f.is_file():
                shutil.copy2(f, dst_cls / f.name)

    for cls in ['cats', 'dogs']:
        src_cls = source_val / cls
        files = [f for f in src_cls.iterdir() if f.is_file()]
        files.sort(key=lambda p: p.name)
        rng = random.Random(SEED)
        rng.shuffle(files)

        half = len(files) // 2
        val_files = files[:half]
        test_files = files[half:]

        for f in val_files:
            shutil.copy2(f, WORKING_SPLIT_DIR / 'validation' / cls / f.name)
        for f in test_files:
            shutil.copy2(f, WORKING_SPLIT_DIR / 'test' / cls / f.name)

    train_dir = WORKING_SPLIT_DIR / 'train'
    validation_dir = WORKING_SPLIT_DIR / 'validation'
    test_dir = WORKING_SPLIT_DIR / 'test'
    split_note = (
        'No dataset-provided test directory was found. Created disjoint␣
↪validation/test '
```

```
        'splits from the provided validation set (500 val + 500 test total).'
    )

print(split_note)
split_df = print_split_table(
    'Dataset split used by this notebook:',
    {'train': train_dir, 'validation': validation_dir, 'test': test_dir}
)

required = {'train': 2000, 'validation': 1000, 'test': 1000}
actual = {r['split']: int(r['total']) for _, r in split_df.iterrows()}
print('Required totals:', required)
print('Actual totals:  ', actual)
if actual != required:
    print(
        '\nNOTE: Current local dataset does not exactly match the required 2000/
↪1000/1000 split. '
        'Results remain valid for this available split, and all comparisons use␣
↪the same split.'
    )
```

```
No dataset-provided test directory was found. Created disjoint validation/test
splits from the provided validation set (500 val + 500 test total).

Dataset split used by this notebook:

        split  cats  dogs  total  \
0       train  1000  1000   2000
1  validation   250   250    500
2        test   250   250    500

                                                   path
0  /home/solyvie/workspace/cs-projects/25263CSDMa…
1  /home/solyvie/workspace/cs-projects/25263CSDMa…
2  /home/solyvie/workspace/cs-projects/25263CSDMa…

Required totals: {'train': 2000, 'validation': 1000, 'test': 1000}
Actual totals:   {'train': 2000, 'validation': 500, 'test': 500}

NOTE: Current local dataset does not exactly match the required 2000/1000/1000
split. Results remain valid for this available split, and all comparisons use
the same split.
```

[53]:
```
IMG_SIZE = (150, 150)
BATCH_SIZE = 20


def make_generators(train_aug_kwargs, batch_size=BATCH_SIZE):
```

```
    train_datagen = ImageDataGenerator(**train_aug_kwargs)
    eval_datagen = ImageDataGenerator(rescale=1.0 / 255)

    train_gen = train_datagen.flow_from_directory(
        str(train_dir),
        target_size=IMG_SIZE,
        batch_size=batch_size,
        class_mode='binary',
        shuffle=True,
        seed=SEED,
    )

    val_gen = eval_datagen.flow_from_directory(
        str(validation_dir),
        target_size=IMG_SIZE,
        batch_size=batch_size,
        class_mode='binary',
        shuffle=False,
    )

    test_gen = eval_datagen.flow_from_directory(
        str(test_dir),
        target_size=IMG_SIZE,
        batch_size=batch_size,
        class_mode='binary',
        shuffle=False,
    )

    return train_gen, val_gen, test_gen


def build_transfer_model(backbone_name='VGG16'):
    if backbone_name == 'VGG16':
        conv_base = VGG16(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3))
        pooling = 'flatten'
    elif backbone_name == 'VGG19':
        conv_base = VGG19(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3))
        pooling = 'flatten'
    elif backbone_name == 'Xception':
        conv_base = Xception(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3))
        pooling = 'gap'
    else:
        raise ValueError(f'Unsupported backbone: {backbone_name}')
```

```python
    conv_base.trainable = False

    model = models.Sequential(name=f'{backbone_name}_transfer')
    model.add(conv_base)
    if pooling == 'flatten':
        model.add(layers.Flatten())
    else:
        model.add(layers.GlobalAveragePooling2D())
    model.add(layers.Dense(256, activation='relu'))
    model.add(layers.Dropout(0.5))
    model.add(layers.Dense(1, activation='sigmoid'))

    model.compile(
        optimizer=optimizers.RMSprop(learning_rate=2e-5),
        loss='binary_crossentropy',
        metrics=['accuracy'],
    )

    return model


def save_h5(model, model_name):
    model_path = OUTPUT_DIR / f'{model_name}.h5'
    model.save(str(model_path))
    return model_path


def plot_history(history, title):
    acc = history.history.get('accuracy', [])
    val_acc = history.history.get('val_accuracy', [])
    loss = history.history.get('loss', [])
    val_loss = history.history.get('val_loss', [])

    epochs = list(range(1, len(acc) + 1))

    fig, axes = plt.subplots(1, 2, figsize=(12, 4))

    axes[0].plot(epochs, acc, label='Train Accuracy')
    axes[0].plot(epochs, val_acc, label='Validation Accuracy')
    axes[0].set_title(f'{title} - Accuracy')
    axes[0].set_xlabel('Epoch')
    axes[0].set_ylabel('Accuracy')
    axes[0].grid(True)
    axes[0].legend()

    axes[1].plot(epochs, loss, label='Train Loss')
    axes[1].plot(epochs, val_loss, label='Validation Loss')
```

```
    axes[1].set_title(f'{title} - Loss')
    axes[1].set_xlabel('Epoch')
    axes[1].set_ylabel('Loss')
    axes[1].grid(True)
    axes[1].legend()

    plt.tight_layout()
    plt.show()


def train_eval_save(
    model_name,
    backbone_name,
    train_aug_kwargs,
    epochs=30,
    batch_size=BATCH_SIZE,
    early_stopping=True,
    steps_per_epoch=None,
    verbose=1,
):
    train_gen, val_gen, test_gen = make_generators(train_aug_kwargs,␣
 ↪batch_size=batch_size)

    model = build_transfer_model(backbone_name)

    callbacks = []
    if early_stopping:
        callbacks.append(
            EarlyStopping(
                monitor='val_loss',
                patience=5,
                restore_best_weights=True,
                verbose=1,
            )
        )

    start = time.time()
    history = model.fit(
        train_gen,
        epochs=epochs,
        validation_data=val_gen,
        callbacks=callbacks,
        steps_per_epoch=steps_per_epoch,
        verbose=verbose,
    )
    train_time = time.time() - start
```

```
    val_loss, val_acc = model.evaluate(val_gen, verbose=0)
    test_loss, test_acc = model.evaluate(test_gen, verbose=0)

    model_path = save_h5(model, model_name)

    result = {
        'model_name': model_name,
        'backbone': backbone_name,
        'epochs_ran': len(history.history['accuracy']),
        'best_val_acc_during_train': float(max(history.
 ↪history['val_accuracy'])),
        'final_train_acc': float(history.history['accuracy'][-1]),
        'final_val_acc': float(val_acc),
        'final_test_acc': float(test_acc),
        'train_time_sec': float(train_time),
        'model_h5_path': str(model_path),
        'history': history,
        'model_obj': model,
    }

    return result
```

1.1 1) Improve Model via `ImageDataGenerator` Hyperparameters

[54]:
```
default_aug = {
    'rescale': 1.0 / 255,
    'rotation_range': 20,
    'width_shift_range': 0.10,
    'height_shift_range': 0.10,
    'shear_range': 0.10,
    'zoom_range': 0.10,
    'horizontal_flip': True,
    'fill_mode': 'nearest',
}

modified_aug = {
    'rescale': 1.0 / 255,
    'rotation_range': 35,
    'width_shift_range': 0.20,
    'height_shift_range': 0.20,
    'shear_range': 0.15,
    'zoom_range': 0.20,
    'horizontal_flip': True,
    'brightness_range': (0.85, 1.15),
    'channel_shift_range': 20.0,
    'fill_mode': 'nearest',
}
```

```python
q1_default = train_eval_save(
    model_name='Q1_VGG16_default_aug',
    backbone_name='VGG16',
    train_aug_kwargs=default_aug,
    epochs=30,
    early_stopping=True,
    verbose=1,
)

q1_modified = train_eval_save(
    model_name='Q1_VGG16_modified_aug',
    backbone_name='VGG16',
    train_aug_kwargs=modified_aug,
    epochs=30,
    early_stopping=True,
    verbose=1,
)

q1_df = pd.DataFrame([
    {k: v for k, v in q1_default.items() if k not in ['history', 'model_obj']},
    {k: v for k, v in q1_modified.items() if k not in ['history', 'model_obj']},
])

q1_display_cols = [
    'model_name', 'epochs_ran', 'best_val_acc_during_train',
    'final_train_acc', 'final_val_acc', 'final_test_acc',
    'train_time_sec', 'model_h5_path'
]

print('Q1 Results Table')
display(q1_df[q1_display_cols].sort_values('final_test_acc', ascending=False).
 ↪reset_index(drop=True))

q1_best = q1_df.sort_values('final_test_acc', ascending=False).iloc[0]
print(
    f"Best Q1 configuration by test accuracy: {q1_best['model_name']} "
    f"(test_acc={q1_best['final_test_acc']:.4f})"
)

plot_history(q1_default['history'], 'Q1 - VGG16 Default Augmentation')
plot_history(q1_modified['history'], 'Q1 - VGG16 Modified Augmentation')
```

```
Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Epoch 1/30
```

```
I0000 00:00:1777522274.451715   23844 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_220068__.62

100/100              20s 140ms/step -
accuracy: 0.6230 - loss: 0.6416 - val_accuracy: 0.8240 - val_loss: 0.4599
Epoch 2/30
100/100              12s 122ms/step -
accuracy: 0.7645 - loss: 0.4927 - val_accuracy: 0.8620 - val_loss: 0.3691
Epoch 3/30
100/100              12s 122ms/step -
accuracy: 0.8100 - loss: 0.4255 - val_accuracy: 0.8700 - val_loss: 0.3288
Epoch 4/30
100/100              13s 132ms/step -
accuracy: 0.8235 - loss: 0.3853 - val_accuracy: 0.8740 - val_loss: 0.3073
Epoch 5/30
100/100              13s 132ms/step -
accuracy: 0.8485 - loss: 0.3568 - val_accuracy: 0.8760 - val_loss: 0.2941
Epoch 6/30
100/100              12s 121ms/step -
accuracy: 0.8395 - loss: 0.3532 - val_accuracy: 0.8880 - val_loss: 0.2843
Epoch 7/30
100/100              12s 121ms/step -
accuracy: 0.8565 - loss: 0.3287 - val_accuracy: 0.8700 - val_loss: 0.2950
Epoch 8/30
100/100              12s 122ms/step -
accuracy: 0.8640 - loss: 0.3158 - val_accuracy: 0.8920 - val_loss: 0.2646
Epoch 9/30
100/100              13s 125ms/step -
accuracy: 0.8685 - loss: 0.3044 - val_accuracy: 0.8960 - val_loss: 0.2588
Epoch 10/30
100/100              12s 116ms/step -
accuracy: 0.8580 - loss: 0.3144 - val_accuracy: 0.8860 - val_loss: 0.2780
Epoch 11/30
100/100              13s 133ms/step -
accuracy: 0.8695 - loss: 0.2950 - val_accuracy: 0.8980 - val_loss: 0.2622
Epoch 12/30
100/100              13s 134ms/step -
accuracy: 0.8810 - loss: 0.2907 - val_accuracy: 0.8980 - val_loss: 0.2525
Epoch 13/30
100/100              11s 113ms/step -
accuracy: 0.8765 - loss: 0.2912 - val_accuracy: 0.8920 - val_loss: 0.2515
Epoch 14/30
100/100              12s 121ms/step -
accuracy: 0.8805 - loss: 0.2836 - val_accuracy: 0.8980 - val_loss: 0.2491
Epoch 15/30
100/100              11s 112ms/step -
accuracy: 0.8845 - loss: 0.2657 - val_accuracy: 0.8960 - val_loss: 0.2466
Epoch 16/30
```

```
100/100              12s 121ms/step -
accuracy: 0.8875 - loss: 0.2695 - val_accuracy: 0.9020 - val_loss: 0.2435
Epoch 17/30
100/100              12s 117ms/step -
accuracy: 0.8915 - loss: 0.2600 - val_accuracy: 0.8960 - val_loss: 0.2447
Epoch 18/30
100/100              11s 107ms/step -
accuracy: 0.8880 - loss: 0.2597 - val_accuracy: 0.8940 - val_loss: 0.2536
Epoch 19/30
100/100              13s 125ms/step -
accuracy: 0.8930 - loss: 0.2499 - val_accuracy: 0.9000 - val_loss: 0.2424
Epoch 20/30
100/100              11s 109ms/step -
accuracy: 0.9025 - loss: 0.2480 - val_accuracy: 0.8980 - val_loss: 0.2424
Epoch 21/30
100/100              13s 125ms/step -
accuracy: 0.8845 - loss: 0.2571 - val_accuracy: 0.9060 - val_loss: 0.2441
Epoch 22/30
100/100              12s 116ms/step -
accuracy: 0.8870 - loss: 0.2464 - val_accuracy: 0.9060 - val_loss: 0.2379
Epoch 23/30
100/100              11s 113ms/step -
accuracy: 0.9015 - loss: 0.2414 - val_accuracy: 0.9080 - val_loss: 0.2361
Epoch 24/30
100/100              12s 117ms/step -
accuracy: 0.8890 - loss: 0.2491 - val_accuracy: 0.8980 - val_loss: 0.2376
Epoch 25/30
100/100              12s 117ms/step -
accuracy: 0.9020 - loss: 0.2374 - val_accuracy: 0.8960 - val_loss: 0.2391
Epoch 26/30
100/100              12s 119ms/step -
accuracy: 0.8910 - loss: 0.2492 - val_accuracy: 0.8980 - val_loss: 0.2373
Epoch 27/30
100/100              12s 120ms/step -
accuracy: 0.8985 - loss: 0.2286 - val_accuracy: 0.9040 - val_loss: 0.2340
Epoch 28/30
100/100              12s 122ms/step -
accuracy: 0.8990 - loss: 0.2347 - val_accuracy: 0.8980 - val_loss: 0.2499
Epoch 29/30
100/100              12s 117ms/step -
accuracy: 0.9015 - loss: 0.2389 - val_accuracy: 0.8980 - val_loss: 0.2388
Epoch 30/30
100/100              12s 120ms/step -
accuracy: 0.9085 - loss: 0.2240 - val_accuracy: 0.8980 - val_loss: 0.2333
Restoring model weights from the end of the best epoch: 30.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
```

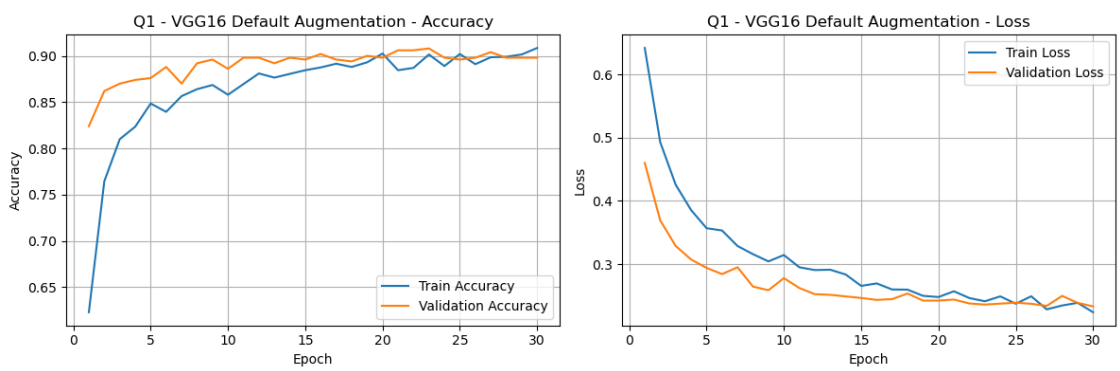
```
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Epoch 1/30

I0000 00:00:1777522649.773588   23844 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_239558__.62

100/100             21s 158ms/step -
accuracy: 0.5845 - loss: 0.6929 - val_accuracy: 0.8120 - val_loss: 0.4914
Epoch 2/30
100/100             14s 135ms/step -
accuracy: 0.6915 - loss: 0.5764 - val_accuracy: 0.8460 - val_loss: 0.4164
Epoch 3/30
100/100             14s 135ms/step -
accuracy: 0.7315 - loss: 0.5201 - val_accuracy: 0.8520 - val_loss: 0.3677
Epoch 4/30
100/100             15s 147ms/step -
accuracy: 0.7680 - loss: 0.4722 - val_accuracy: 0.8500 - val_loss: 0.3335
Epoch 5/30
100/100             14s 137ms/step -
accuracy: 0.7945 - loss: 0.4430 - val_accuracy: 0.8600 - val_loss: 0.3262
Epoch 6/30
100/100             13s 131ms/step -
accuracy: 0.8030 - loss: 0.4314 - val_accuracy: 0.8540 - val_loss: 0.3159
Epoch 7/30
100/100             14s 135ms/step -
accuracy: 0.8075 - loss: 0.4251 - val_accuracy: 0.8720 - val_loss: 0.2897
Epoch 8/30
100/100             13s 132ms/step -
accuracy: 0.8150 - loss: 0.4116 - val_accuracy: 0.8720 - val_loss: 0.2843
Epoch 9/30
100/100             13s 131ms/step -
accuracy: 0.8270 - loss: 0.3965 - val_accuracy: 0.8820 - val_loss: 0.2764
Epoch 10/30
100/100             14s 138ms/step -
accuracy: 0.8230 - loss: 0.3912 - val_accuracy: 0.8780 - val_loss: 0.2742
Epoch 11/30
100/100             14s 137ms/step -
accuracy: 0.8275 - loss: 0.3750 - val_accuracy: 0.8980 - val_loss: 0.2666
Epoch 12/30
100/100             13s 131ms/step -
accuracy: 0.8310 - loss: 0.3730 - val_accuracy: 0.8760 - val_loss: 0.2632
Epoch 13/30
100/100             13s 130ms/step -
```

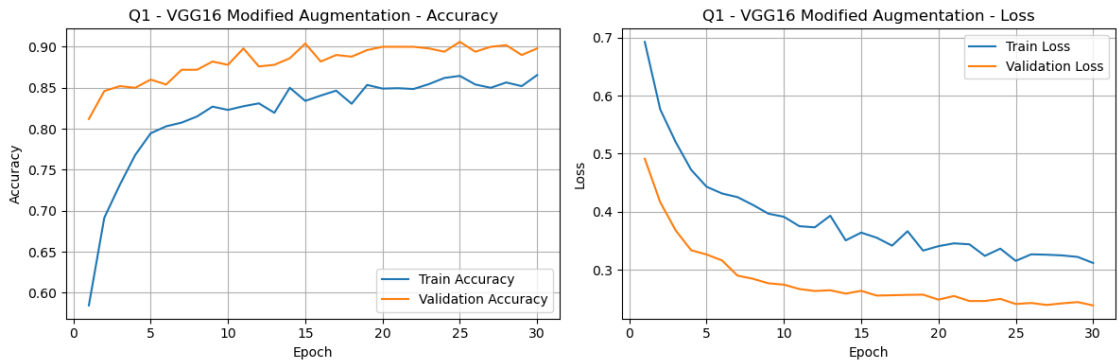
```
accuracy: 0.8195 - loss: 0.3931 - val_accuracy: 0.8780 - val_loss: 0.2645
Epoch 14/30
100/100             13s 127ms/step -
accuracy: 0.8500 - loss: 0.3505 - val_accuracy: 0.8860 - val_loss: 0.2587
Epoch 15/30
100/100             14s 135ms/step -
accuracy: 0.8340 - loss: 0.3639 - val_accuracy: 0.9040 - val_loss: 0.2635
Epoch 16/30
100/100             13s 127ms/step -
accuracy: 0.8405 - loss: 0.3551 - val_accuracy: 0.8820 - val_loss: 0.2553
Epoch 17/30
100/100             12s 118ms/step -
accuracy: 0.8465 - loss: 0.3414 - val_accuracy: 0.8900 - val_loss: 0.2559
Epoch 18/30
100/100             13s 130ms/step -
accuracy: 0.8305 - loss: 0.3663 - val_accuracy: 0.8880 - val_loss: 0.2565
Epoch 19/30
100/100             14s 136ms/step -
accuracy: 0.8535 - loss: 0.3329 - val_accuracy: 0.8960 - val_loss: 0.2569
Epoch 20/30
100/100             13s 131ms/step -
accuracy: 0.8490 - loss: 0.3406 - val_accuracy: 0.9000 - val_loss: 0.2483
Epoch 21/30
100/100             13s 130ms/step -
accuracy: 0.8495 - loss: 0.3454 - val_accuracy: 0.9000 - val_loss: 0.2546
Epoch 22/30
100/100             13s 129ms/step -
accuracy: 0.8485 - loss: 0.3437 - val_accuracy: 0.9000 - val_loss: 0.2460
Epoch 23/30
100/100             14s 136ms/step -
accuracy: 0.8545 - loss: 0.3238 - val_accuracy: 0.8980 - val_loss: 0.2459
Epoch 24/30
100/100             13s 128ms/step -
accuracy: 0.8620 - loss: 0.3362 - val_accuracy: 0.8940 - val_loss: 0.2496
Epoch 25/30
100/100             13s 133ms/step -
accuracy: 0.8645 - loss: 0.3152 - val_accuracy: 0.9060 - val_loss: 0.2408
Epoch 26/30
100/100             13s 130ms/step -
accuracy: 0.8540 - loss: 0.3265 - val_accuracy: 0.8940 - val_loss: 0.2426
Epoch 27/30
100/100             13s 130ms/step -
accuracy: 0.8500 - loss: 0.3258 - val_accuracy: 0.9000 - val_loss: 0.2391
Epoch 28/30
100/100             14s 136ms/step -
accuracy: 0.8565 - loss: 0.3246 - val_accuracy: 0.9020 - val_loss: 0.2419
Epoch 29/30
100/100             14s 137ms/step -
```

```
accuracy: 0.8520 - loss: 0.3220 - val_accuracy: 0.8900 - val_loss: 0.2441
Epoch 30/30
100/100              13s 133ms/step -
accuracy: 0.8655 - loss: 0.3116 - val_accuracy: 0.8980 - val_loss: 0.2383
Restoring model weights from the end of the best epoch: 30.
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.
```

```
Q1 Results Table

              model_name  epochs_ran  best_val_acc_during_train  \
0   Q1_VGG16_default_aug          30                      0.908
1  Q1_VGG16_modified_aug          30                      0.906

   final_train_acc  final_val_acc  final_test_acc  train_time_sec  \
0           0.9085          0.898           0.900      372.180018
1           0.8655          0.898           0.884      408.268546

                                        model_h5_path
0  /home/solyvie/workspace/cs-projects/25263CSDMa…
1  /home/solyvie/workspace/cs-projects/25263CSDMa…

Best Q1 configuration by test accuracy: Q1_VGG16_default_aug (test_acc=0.9000)
```





1.2 2) Implement VGG19 and Compare with VGG16

Both are trained under the same augmentation setup for fair comparison.

[55]:
```
best_aug_name = 'modified_aug' if q1_modified['final_test_acc'] >=␣
↪q1_default['final_test_acc'] else 'default_aug'
best_aug = modified_aug if best_aug_name == 'modified_aug' else default_aug
print('Using augmentation for Q2:', best_aug_name)

q2_vgg16 = train_eval_save(
    model_name='Q2_VGG16',
    backbone_name='VGG16',
    train_aug_kwargs=best_aug,
    epochs=30,
    early_stopping=True,
    verbose=1,
)

q2_vgg19 = train_eval_save(
    model_name='Q2_VGG19',
    backbone_name='VGG19',
    train_aug_kwargs=best_aug,
    epochs=30,
    early_stopping=True,
    verbose=1,
)

arch_stats = []
for backbone in ['VGG16', 'VGG19']:
    if backbone == 'VGG16':
        base = VGG16(weights='imagenet', include_top=False, input_shape=(150,␣
↪150, 3))
    else:
```

```
        base = VGG19(weights='imagenet', include_top=False, input_shape=(150,␣
↪150, 3))
    arch_stats.append({
        'backbone': backbone,
        'num_layers': len(base.layers),
        'num_params': base.count_params(),
    })

arch_df = pd.DataFrame(arch_stats)
q2_df = pd.DataFrame([
    {k: v for k, v in q2_vgg16.items() if k not in ['history', 'model_obj']},
    {k: v for k, v in q2_vgg19.items() if k not in ['history', 'model_obj']},
])
q2_df = q2_df.merge(arch_df, left_on='backbone', right_on='backbone',␣
↪how='left')

q2_display_cols = [
    'model_name', 'backbone', 'num_layers', 'num_params',
    'epochs_ran', 'final_train_acc', 'final_val_acc', 'final_test_acc',
    'train_time_sec', 'model_h5_path'
]

print('Q2 Results Table')
display(q2_df[q2_display_cols].sort_values('final_test_acc', ascending=False).
↪reset_index(drop=True))

q2_best = q2_df.sort_values('final_test_acc', ascending=False).iloc[0]
print(
    f"Best in Q2 by test accuracy: {q2_best['backbone']} "
    f"(test_acc={q2_best['final_test_acc']:.4f})"
)

plot_history(q2_vgg16['history'], 'Q2 - VGG16')
plot_history(q2_vgg19['history'], 'Q2 - VGG19')
```

```
Using augmentation for Q2: default_aug
Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Epoch 1/30

I0000 00:00:1777523065.445367   23841 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_259168__.62

100/100              18s 134ms/step -
accuracy: 0.6510 - loss: 0.6270 - val_accuracy: 0.8300 - val_loss: 0.4425
Epoch 2/30
100/100              12s 114ms/step -
```

```
accuracy: 0.7550 - loss: 0.4939 - val_accuracy: 0.8580 - val_loss: 0.3666
Epoch 3/30
100/100             12s 117ms/step -
accuracy: 0.8020 - loss: 0.4260 - val_accuracy: 0.8700 - val_loss: 0.3267
Epoch 4/30
100/100             12s 117ms/step -
accuracy: 0.8215 - loss: 0.3923 - val_accuracy: 0.8760 - val_loss: 0.3066
Epoch 5/30
100/100             12s 119ms/step -
accuracy: 0.8375 - loss: 0.3743 - val_accuracy: 0.8820 - val_loss: 0.2937
Epoch 6/30
100/100             12s 120ms/step -
accuracy: 0.8405 - loss: 0.3525 - val_accuracy: 0.8980 - val_loss: 0.2779
Epoch 7/30
100/100             12s 121ms/step -
accuracy: 0.8460 - loss: 0.3479 - val_accuracy: 0.8900 - val_loss: 0.2718
Epoch 8/30
100/100             12s 115ms/step -
accuracy: 0.8710 - loss: 0.3251 - val_accuracy: 0.8880 - val_loss: 0.2720
Epoch 9/30
100/100             12s 120ms/step -
accuracy: 0.8700 - loss: 0.3035 - val_accuracy: 0.8940 - val_loss: 0.2645
Epoch 10/30
100/100             12s 115ms/step -
accuracy: 0.8635 - loss: 0.3081 - val_accuracy: 0.8980 - val_loss: 0.2577
Epoch 11/30
100/100             12s 116ms/step -
accuracy: 0.8665 - loss: 0.3070 - val_accuracy: 0.8980 - val_loss: 0.2560
Epoch 12/30
100/100             20s 113ms/step -
accuracy: 0.8735 - loss: 0.2961 - val_accuracy: 0.8980 - val_loss: 0.2546
Epoch 13/30
100/100             13s 126ms/step -
accuracy: 0.8780 - loss: 0.2868 - val_accuracy: 0.9020 - val_loss: 0.2497
Epoch 14/30
100/100             13s 125ms/step -
accuracy: 0.8810 - loss: 0.2833 - val_accuracy: 0.9100 - val_loss: 0.2509
Epoch 15/30
100/100             13s 126ms/step -
accuracy: 0.8880 - loss: 0.2756 - val_accuracy: 0.9100 - val_loss: 0.2475
Epoch 16/30
100/100             14s 136ms/step -
accuracy: 0.8815 - loss: 0.2717 - val_accuracy: 0.9120 - val_loss: 0.2449
Epoch 17/30
100/100             13s 133ms/step -
accuracy: 0.8855 - loss: 0.2655 - val_accuracy: 0.9120 - val_loss: 0.2429
Epoch 18/30
100/100             12s 120ms/step -
```

```
accuracy: 0.8905 - loss: 0.2636 - val_accuracy: 0.9080 - val_loss: 0.2413
Epoch 19/30
100/100             12s 122ms/step -
accuracy: 0.8955 - loss: 0.2548 - val_accuracy: 0.9080 - val_loss: 0.2405
Epoch 20/30
100/100             12s 119ms/step -
accuracy: 0.8885 - loss: 0.2583 - val_accuracy: 0.9060 - val_loss: 0.2368
Epoch 21/30
100/100             13s 126ms/step -
accuracy: 0.8985 - loss: 0.2432 - val_accuracy: 0.9140 - val_loss: 0.2384
Epoch 22/30
100/100             12s 121ms/step -
accuracy: 0.8890 - loss: 0.2607 - val_accuracy: 0.9040 - val_loss: 0.2355
Epoch 23/30
100/100             12s 120ms/step -
accuracy: 0.9020 - loss: 0.2327 - val_accuracy: 0.9080 - val_loss: 0.2495
Epoch 24/30
100/100             12s 120ms/step -
accuracy: 0.9010 - loss: 0.2366 - val_accuracy: 0.9140 - val_loss: 0.2339
Epoch 25/30
100/100             12s 124ms/step -
accuracy: 0.9020 - loss: 0.2399 - val_accuracy: 0.9120 - val_loss: 0.2331
Epoch 26/30
100/100             12s 123ms/step -
accuracy: 0.8980 - loss: 0.2399 - val_accuracy: 0.9140 - val_loss: 0.2354
Epoch 27/30
100/100             12s 119ms/step -
accuracy: 0.8935 - loss: 0.2441 - val_accuracy: 0.8960 - val_loss: 0.2537
Epoch 28/30
100/100             13s 126ms/step -
accuracy: 0.9040 - loss: 0.2319 - val_accuracy: 0.9060 - val_loss: 0.2320
Epoch 29/30
100/100             12s 116ms/step -
accuracy: 0.9000 - loss: 0.2366 - val_accuracy: 0.9140 - val_loss: 0.2352
Epoch 30/30
100/100             12s 117ms/step -
accuracy: 0.9030 - loss: 0.2374 - val_accuracy: 0.9100 - val_loss: 0.2305
Restoring model weights from the end of the best epoch: 30.
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.
```

```
Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Epoch 1/30
```

```
I0000 00:00:1777523448.417399   23844 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_279150__.65

100/100              18s 141ms/step -
accuracy: 0.6145 - loss: 0.6559 - val_accuracy: 0.7920 - val_loss: 0.4996
Epoch 2/30
100/100              12s 116ms/step -
accuracy: 0.7245 - loss: 0.5387 - val_accuracy: 0.8100 - val_loss: 0.4371
Epoch 3/30
100/100              12s 122ms/step -
accuracy: 0.7650 - loss: 0.4838 - val_accuracy: 0.8360 - val_loss: 0.3940
Epoch 4/30
100/100              13s 124ms/step -
accuracy: 0.7935 - loss: 0.4447 - val_accuracy: 0.8360 - val_loss: 0.3723
Epoch 5/30
100/100              14s 136ms/step -
accuracy: 0.8120 - loss: 0.4096 - val_accuracy: 0.8440 - val_loss: 0.3551
Epoch 6/30
100/100              12s 123ms/step -
accuracy: 0.8050 - loss: 0.4066 - val_accuracy: 0.8680 - val_loss: 0.3395
Epoch 7/30
100/100              12s 120ms/step -
accuracy: 0.8280 - loss: 0.3829 - val_accuracy: 0.8620 - val_loss: 0.3341
Epoch 8/30
100/100              12s 116ms/step -
accuracy: 0.8290 - loss: 0.3824 - val_accuracy: 0.8640 - val_loss: 0.3263
Epoch 9/30
100/100              12s 122ms/step -
accuracy: 0.8295 - loss: 0.3683 - val_accuracy: 0.8640 - val_loss: 0.3216
Epoch 10/30
100/100              21s 120ms/step -
accuracy: 0.8370 - loss: 0.3669 - val_accuracy: 0.8560 - val_loss: 0.3318
Epoch 11/30
100/100              12s 118ms/step -
accuracy: 0.8510 - loss: 0.3464 - val_accuracy: 0.8700 - val_loss: 0.3205
Epoch 12/30
100/100              12s 123ms/step -
accuracy: 0.8525 - loss: 0.3346 - val_accuracy: 0.8600 - val_loss: 0.3270
Epoch 13/30
100/100              13s 126ms/step -
accuracy: 0.8545 - loss: 0.3409 - val_accuracy: 0.8480 - val_loss: 0.3363
Epoch 14/30
100/100              13s 129ms/step -
accuracy: 0.8580 - loss: 0.3262 - val_accuracy: 0.8680 - val_loss: 0.3194
Epoch 15/30
100/100              13s 125ms/step -
accuracy: 0.8585 - loss: 0.3266 - val_accuracy: 0.8720 - val_loss: 0.3140
Epoch 16/30
```

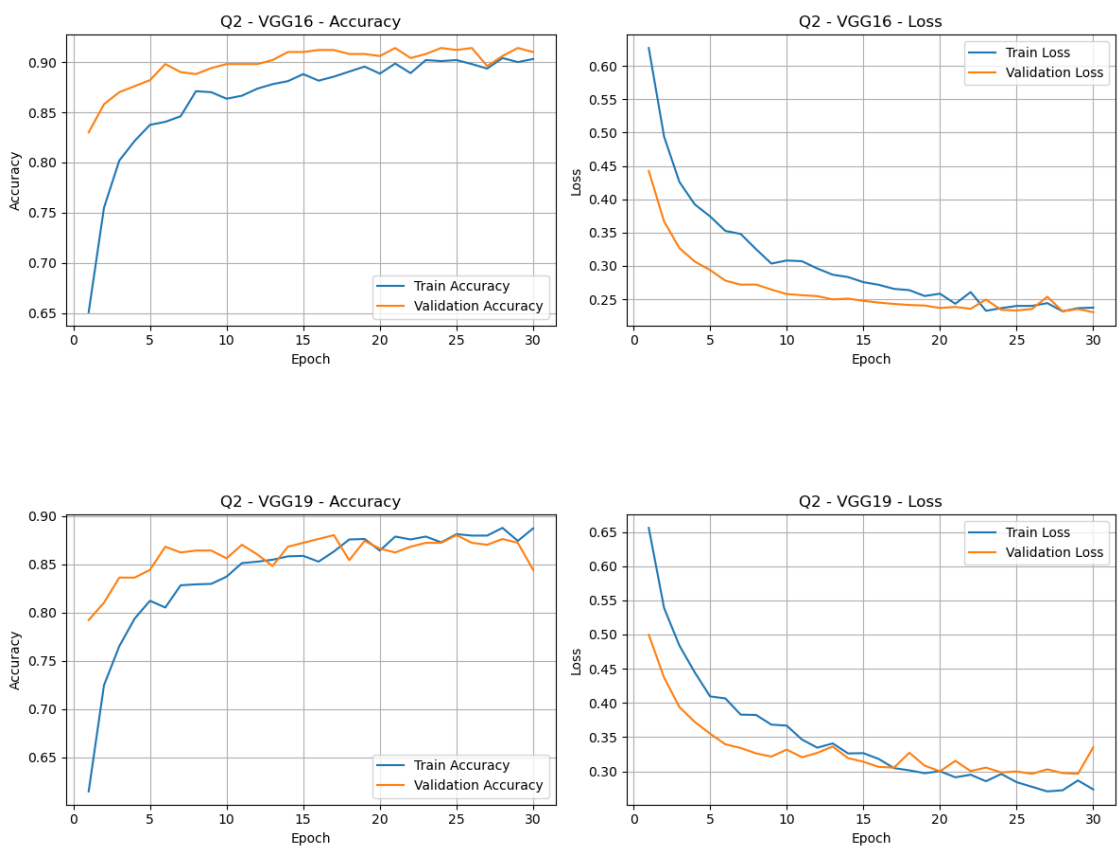
```
100/100              12s 122ms/step -
accuracy: 0.8525 - loss: 0.3181 - val_accuracy: 0.8760 - val_loss: 0.3066
Epoch 17/30
100/100              12s 122ms/step -
accuracy: 0.8630 - loss: 0.3047 - val_accuracy: 0.8800 - val_loss: 0.3054
Epoch 18/30
100/100              12s 115ms/step -
accuracy: 0.8755 - loss: 0.3015 - val_accuracy: 0.8540 - val_loss: 0.3272
Epoch 19/30
100/100              13s 126ms/step -
accuracy: 0.8760 - loss: 0.2973 - val_accuracy: 0.8740 - val_loss: 0.3083
Epoch 20/30
100/100              12s 123ms/step -
accuracy: 0.8640 - loss: 0.3002 - val_accuracy: 0.8660 - val_loss: 0.3001
Epoch 21/30
100/100              12s 119ms/step -
accuracy: 0.8785 - loss: 0.2913 - val_accuracy: 0.8620 - val_loss: 0.3154
Epoch 22/30
100/100              12s 117ms/step -
accuracy: 0.8755 - loss: 0.2951 - val_accuracy: 0.8680 - val_loss: 0.3004
Epoch 23/30
100/100              13s 128ms/step -
accuracy: 0.8785 - loss: 0.2855 - val_accuracy: 0.8720 - val_loss: 0.3055
Epoch 24/30
100/100              13s 126ms/step -
accuracy: 0.8725 - loss: 0.2962 - val_accuracy: 0.8720 - val_loss: 0.2984
Epoch 25/30
100/100              12s 119ms/step -
accuracy: 0.8810 - loss: 0.2842 - val_accuracy: 0.8800 - val_loss: 0.2998
Epoch 26/30
100/100              12s 123ms/step -
accuracy: 0.8795 - loss: 0.2774 - val_accuracy: 0.8720 - val_loss: 0.2964
Epoch 27/30
100/100              12s 123ms/step -
accuracy: 0.8795 - loss: 0.2707 - val_accuracy: 0.8700 - val_loss: 0.3029
Epoch 28/30
100/100              13s 126ms/step -
accuracy: 0.8875 - loss: 0.2723 - val_accuracy: 0.8760 - val_loss: 0.2975
Epoch 29/30
100/100              13s 126ms/step -
accuracy: 0.8740 - loss: 0.2867 - val_accuracy: 0.8720 - val_loss: 0.2964
Epoch 30/30
100/100              12s 118ms/step -
accuracy: 0.8870 - loss: 0.2736 - val_accuracy: 0.8440 - val_loss: 0.3351
Restoring model weights from the end of the best epoch: 26.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
```

```
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.
```

Q2 Results Table

```
  model_name backbone  num_layers  num_params  epochs_ran  final_train_acc  \
0   Q2_VGG19    VGG19          22    20024384          30            0.887
1   Q2_VGG16    VGG16          19    14714688          30            0.903

   final_val_acc  final_test_acc  train_time_sec  \
0          0.872           0.910      384.580479
1          0.910           0.898      379.770050

                                       model_h5_path
0  /home/solyvie/workspace/cs-projects/25263CSDMa…
1  /home/solyvie/workspace/cs-projects/25263CSDMa…
```

Best in Q2 by test accuracy: VGG19 (test_acc=0.9100)



1.3 3) VGG16 Test Accuracy vs Training Samples (No Data Augmentation)

Required sample sizes: `500, 1000, 2000, 5000, 10000, 15000`. Each run uses **30 epochs**.

Note: Q3 training is run with `verbose=0` to suppress epoch logs and keep PDF export compact.

[56]:
```
sample_sizes = [500, 1000, 2000, 5000, 10000, 15000]
q3_results = []
q3_full_runs = []

no_aug = {'rescale': 1.0 / 255}

for n_samples in sample_sizes:
    steps = max(1, int(np.ceil(n_samples / BATCH_SIZE)))
    run_name = f'Q3_VGG16_noaug_{n_samples}samples'

    run = train_eval_save(
        model_name=run_name,
        backbone_name='VGG16',
        train_aug_kwargs=no_aug,
        epochs=30,
        early_stopping=True,
        steps_per_epoch=steps,
        verbose=0,
    )

    q3_full_runs.append(run)
    q3_results.append({
        'samples_per_epoch_target': n_samples,
        'steps_per_epoch': steps,
        'epochs_ran': run['epochs_ran'],
        'final_train_acc': run['final_train_acc'],
        'final_val_acc': run['final_val_acc'],
        'final_test_acc': run['final_test_acc'],
        'train_time_sec': run['train_time_sec'],
        'model_h5_path': run['model_h5_path'],
    })

q3_df = pd.DataFrame(q3_results)
print('Q3 Results Table')
display(q3_df)

plt.figure(figsize=(9, 5))
plt.plot(q3_df['samples_per_epoch_target'], q3_df['final_test_acc'],␣
 ↪marker='o', label='Test Accuracy')
plt.plot(q3_df['samples_per_epoch_target'], q3_df['final_val_acc'], marker='o',␣
 ↪label='Validation Accuracy')
plt.xlabel('Training samples per epoch (target)')
plt.ylabel('Accuracy')
plt.title('VGG16 Accuracy vs Training Sample Budget (No Augmentation, 30␣
 ↪Epochs)')
```

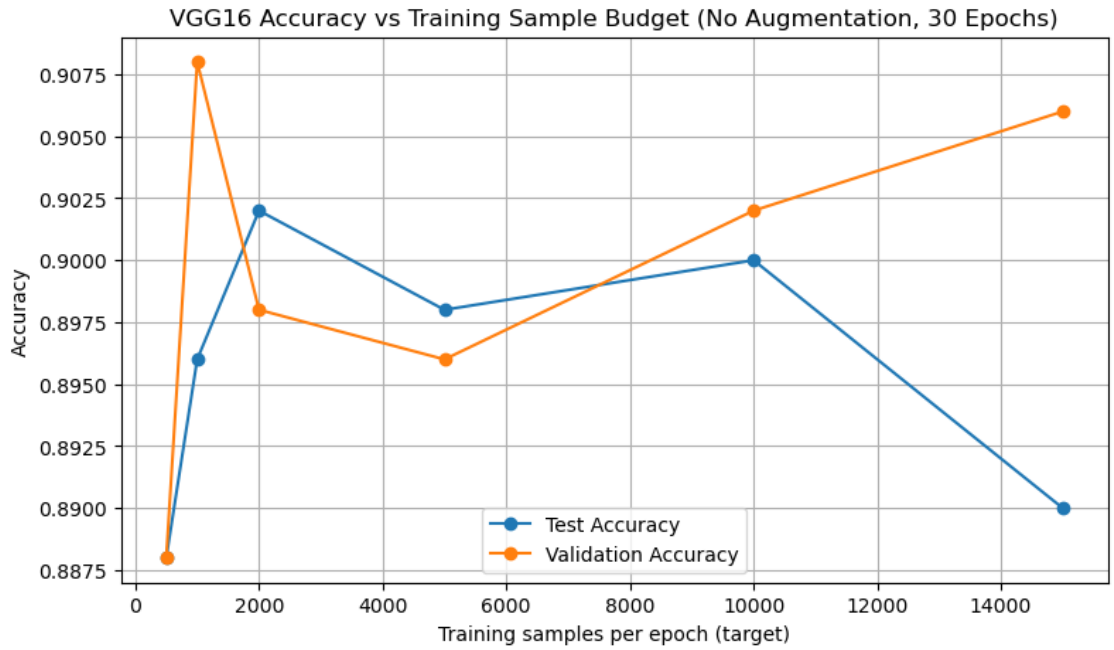
```python
plt.grid(True)
plt.legend()
plt.show()

best_q3 = q3_df.sort_values('final_test_acc', ascending=False).iloc[0]
print(
    f"Best Q3 run: {int(best_q3['samples_per_epoch_target'])} samples/epoch "
    f"(test_acc={best_q3['final_test_acc']:.4f})"
)

for run in q3_full_runs:
    plot_history(run['history'], f"Q3 - {run['model_name']}")
```

```
Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.

I0000 00:00:1777523837.259918   23845 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_299564__.62

Restoring model weights from the end of the best epoch: 30.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.

I0000 00:00:1777523904.687969   23843 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_310422__.62

Epoch 29: early stopping
Restoring model weights from the end of the best epoch: 24.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.

I0000 00:00:1777524000.780493   23841 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_323782__.62

Epoch 18: early stopping
Restoring model weights from the end of the best epoch: 13.
```

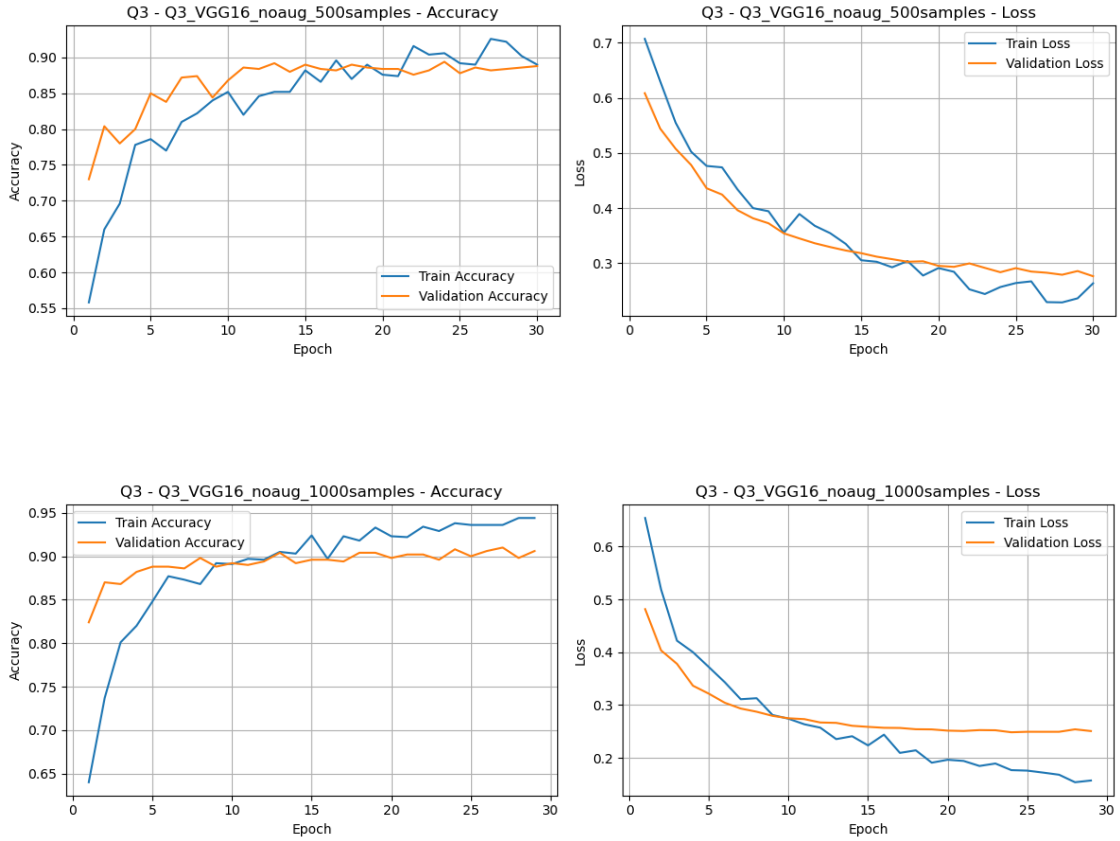
```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.

I0000 00:00:1777524095.880570   23845 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_336631__.62
/home/solyvie/environments/miniforge3/envs/thesis-env/lib/python3.13/site-
packages/keras/src/trainers/epoch_iterator.py:116: UserWarning: Your input ran
out of data; interrupting training. Make sure that your dataset or generator can
generate at least `steps_per_epoch * epochs` batches. You may need to use the
`.repeat()` function when building your dataset.
  self._interrupted_warning()

Epoch 20: early stopping
Restoring model weights from the end of the best epoch: 15.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.

I0000 00:00:1777524199.324901   23841 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_350472__.62

Epoch 20: early stopping
Restoring model weights from the end of the best epoch: 15.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.

I0000 00:00:1777524302.963603   23845 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_364433__.62

Epoch 19: early stopping
Restoring model weights from the end of the best epoch: 14.
```

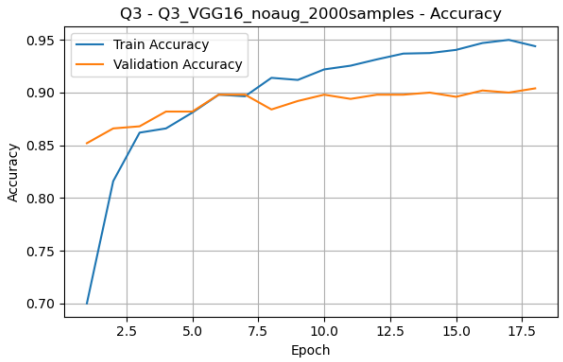
```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.
```

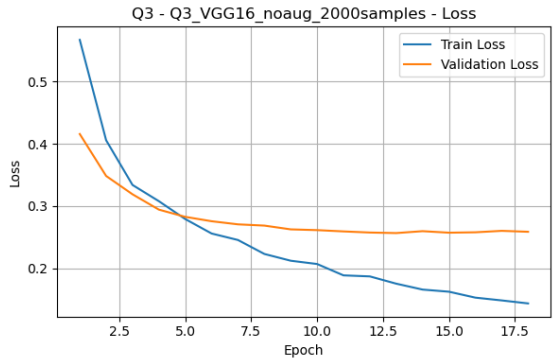
```
Q3 Results Table

   samples_per_epoch_target  steps_per_epoch  epochs_ran  final_train_acc  \
0                       500               25          30           0.8900
1                      1000               50          29           0.9440
2                      2000              100          18           0.9440
3                      5000              250          20           0.9530
4                     10000              500          20           0.9520
5                     15000              750          19           0.9535

   final_val_acc  final_test_acc  train_time_sec  \
0          0.888           0.888       64.257334
1          0.908           0.896       91.852459
2          0.898           0.902       92.207176
3          0.896           0.898      100.421194
4          0.902           0.900      101.510481
5          0.906           0.890       96.281361

                                      model_h5_path
0  /home/solyvie/workspace/cs-projects/25263CSDMa…
1  /home/solyvie/workspace/cs-projects/25263CSDMa…
2  /home/solyvie/workspace/cs-projects/25263CSDMa…
3  /home/solyvie/workspace/cs-projects/25263CSDMa…
4  /home/solyvie/workspace/cs-projects/25263CSDMa…
5  /home/solyvie/workspace/cs-projects/25263CSDMa…
```

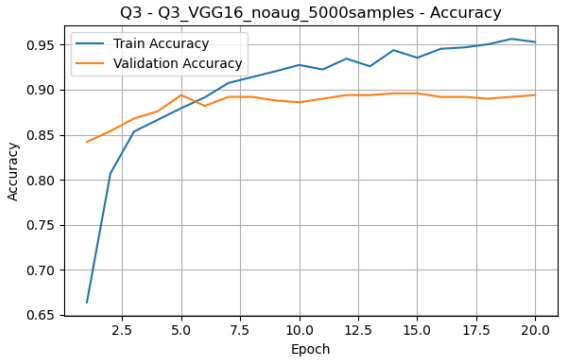


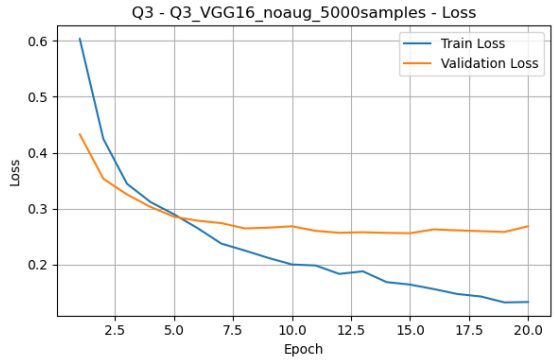

```
Best Q3 run: 2000 samples/epoch (test_acc=0.9020)
```

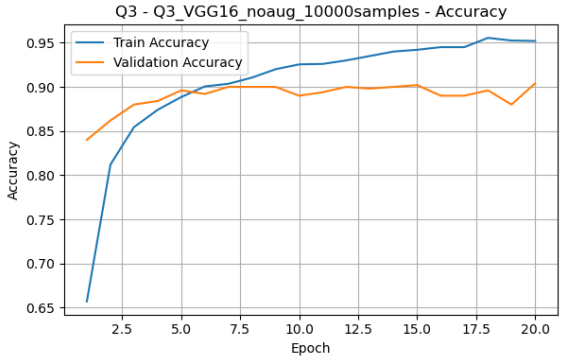


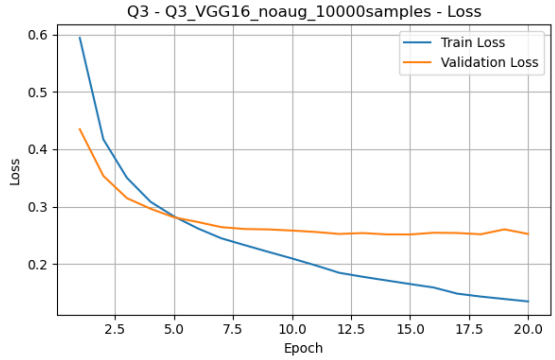

Q3 - Q3_VGG16_noaug_2000samples - Accuracy
Train Accuracy
Validation Accuracy
Accuracy
Epoch

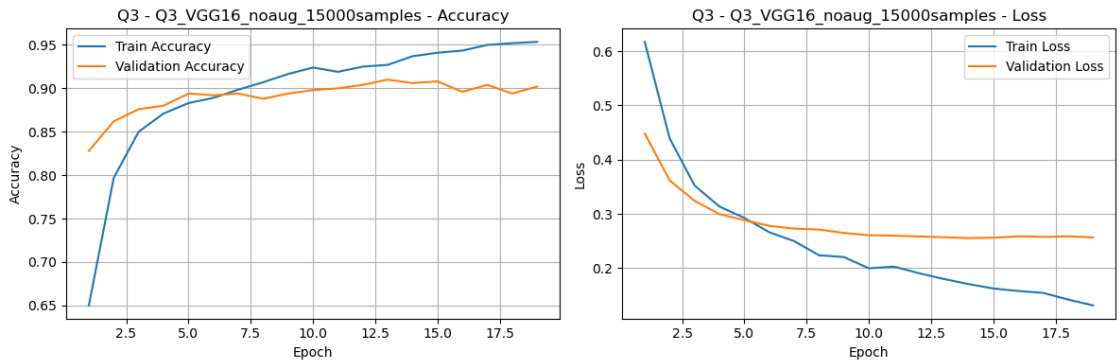

Q3 - Q3_VGG16_noaug_2000samples - Loss
Train Loss
Validation Loss
Loss
Epoch


Q3 - Q3_VGG16_noaug_5000samples - Accuracy
Train Accuracy
Validation Accuracy
Accuracy
Epoch


Q3 - Q3_VGG16_noaug_5000samples - Loss
Train Loss
Validation Loss
Loss
Epoch


Q3 - Q3_VGG16_noaug_10000samples - Accuracy
Train Accuracy
Validation Accuracy
Accuracy
Epoch


Q3 - Q3_VGG16_noaug_10000samples - Loss
Train Loss
Validation Loss
Loss
Epoch



1.4 4) Implement Xception and Compare Architecture + Accuracy

Uses the same train/validation/test split configured at the start of this notebook.

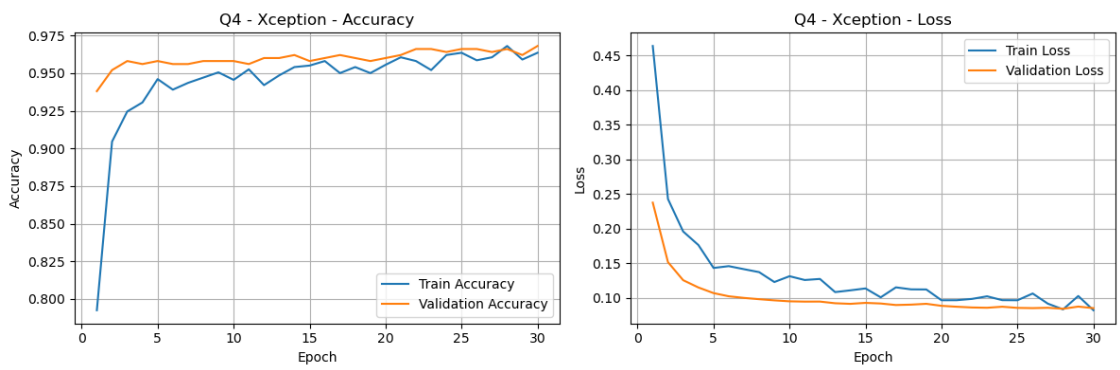
[57]:
```
q4_xception = train_eval_save(
    model_name='Q4_Xception',
    backbone_name='Xception',
    train_aug_kwargs=best_aug,
    epochs=30,
    early_stopping=True,
    verbose=1,
)

q1_best_row = q1_df.sort_values('final_test_acc', ascending=False).iloc[0]
q2_best_row = q2_df.sort_values('final_test_acc', ascending=False).iloc[0]

arch_meta = {
    'VGG16': {
        'num_layers': len(VGG16(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3)).layers),
        'num_params': VGG16(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3)).count_params(),
    },
    'VGG19': {
        'num_layers': len(VGG19(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3)).layers),
        'num_params': VGG19(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3)).count_params(),
    },
    'Xception': {
        'num_layers': len(Xception(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3)).layers),
```

```
        'num_params': Xception(weights='imagenet', include_top=False,␣
 ↪input_shape=(150, 150, 3)).count_params(),
    },
}

q4_compare = pd.DataFrame([
    {
        'model_type': 'Best VGG16/VGG19 from Q2',
        'backbone': q2_best_row['backbone'],
        'num_layers': arch_meta[q2_best_row['backbone']]['num_layers'],
        'num_params': arch_meta[q2_best_row['backbone']]['num_params'],
        'final_val_acc': q2_best_row['final_val_acc'],
        'final_test_acc': q2_best_row['final_test_acc'],
        'model_h5_path': q2_best_row['model_h5_path'],
    },
    {
        'model_type': 'Best augmentation VGG16 from Q1',
        'backbone': 'VGG16',
        'num_layers': arch_meta['VGG16']['num_layers'],
        'num_params': arch_meta['VGG16']['num_params'],
        'final_val_acc': q1_best_row['final_val_acc'],
        'final_test_acc': q1_best_row['final_test_acc'],
        'model_h5_path': q1_best_row['model_h5_path'],
    },
    {
        'model_type': 'Xception (Q4)',
        'backbone': 'Xception',
        'num_layers': arch_meta['Xception']['num_layers'],
        'num_params': arch_meta['Xception']['num_params'],
        'final_val_acc': q4_xception['final_val_acc'],
        'final_test_acc': q4_xception['final_test_acc'],
        'model_h5_path': q4_xception['model_h5_path'],
    },
])

print('Q4 Comparison Table')
display(q4_compare.sort_values('final_test_acc', ascending=False).
 ↪reset_index(drop=True))

q4_best = q4_compare.sort_values('final_test_acc', ascending=False).iloc[0]
print(
    f"Best overall in Q4 comparison: {q4_best['backbone']} "
    f"(test_acc={q4_best['final_test_acc']:.4f})"
)

plot_history(q4_xception['history'], 'Q4 - Xception')
```

```
Found 2000 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Found 500 images belonging to 2 classes.
Epoch 1/30

I0000 00:00:1777524413.493501   23843 dot_merger.cc:481] Merging Dots in
computation: a_inference_one_step_on_data_383626__.123

100/100             29s 189ms/step -
accuracy: 0.7925 - loss: 0.4639 - val_accuracy: 0.9380 - val_loss: 0.2374
Epoch 2/30
100/100             13s 133ms/step -
accuracy: 0.9045 - loss: 0.2426 - val_accuracy: 0.9520 - val_loss: 0.1513
Epoch 3/30
100/100             12s 122ms/step -
accuracy: 0.9245 - loss: 0.1956 - val_accuracy: 0.9580 - val_loss: 0.1252
Epoch 4/30
100/100             13s 128ms/step -
accuracy: 0.9305 - loss: 0.1761 - val_accuracy: 0.9560 - val_loss: 0.1147
Epoch 5/30
100/100             12s 120ms/step -
accuracy: 0.9460 - loss: 0.1429 - val_accuracy: 0.9580 - val_loss: 0.1066
Epoch 6/30
100/100             11s 108ms/step -
accuracy: 0.9390 - loss: 0.1455 - val_accuracy: 0.9560 - val_loss: 0.1020
Epoch 7/30
100/100             12s 118ms/step -
accuracy: 0.9435 - loss: 0.1412 - val_accuracy: 0.9560 - val_loss: 0.0996
Epoch 8/30
100/100             12s 118ms/step -
accuracy: 0.9470 - loss: 0.1369 - val_accuracy: 0.9580 - val_loss: 0.0977
Epoch 9/30
100/100             13s 127ms/step -
accuracy: 0.9505 - loss: 0.1226 - val_accuracy: 0.9580 - val_loss: 0.0961
Epoch 10/30
100/100             13s 129ms/step -
accuracy: 0.9455 - loss: 0.1310 - val_accuracy: 0.9580 - val_loss: 0.0947
Epoch 11/30
100/100             13s 132ms/step -
accuracy: 0.9525 - loss: 0.1255 - val_accuracy: 0.9560 - val_loss: 0.0942
Epoch 12/30
100/100             12s 118ms/step -
accuracy: 0.9420 - loss: 0.1271 - val_accuracy: 0.9600 - val_loss: 0.0943
Epoch 13/30
100/100             13s 127ms/step -
accuracy: 0.9485 - loss: 0.1080 - val_accuracy: 0.9600 - val_loss: 0.0919
Epoch 14/30
100/100             12s 118ms/step -
accuracy: 0.9540 - loss: 0.1106 - val_accuracy: 0.9620 - val_loss: 0.0910
```

```
Epoch 15/30
100/100              14s 135ms/step -
accuracy: 0.9550 - loss: 0.1133 - val_accuracy: 0.9580 - val_loss: 0.0924
Epoch 16/30
100/100              13s 132ms/step -
accuracy: 0.9580 - loss: 0.1004 - val_accuracy: 0.9600 - val_loss: 0.0915
Epoch 17/30
100/100              12s 121ms/step -
accuracy: 0.9500 - loss: 0.1149 - val_accuracy: 0.9620 - val_loss: 0.0893
Epoch 18/30
100/100              13s 127ms/step -
accuracy: 0.9540 - loss: 0.1120 - val_accuracy: 0.9600 - val_loss: 0.0899
Epoch 19/30
100/100              12s 119ms/step -
accuracy: 0.9500 - loss: 0.1118 - val_accuracy: 0.9580 - val_loss: 0.0911
Epoch 20/30
100/100              13s 129ms/step -
accuracy: 0.9555 - loss: 0.0962 - val_accuracy: 0.9600 - val_loss: 0.0882
Epoch 21/30
100/100              14s 142ms/step -
accuracy: 0.9605 - loss: 0.0963 - val_accuracy: 0.9620 - val_loss: 0.0868
Epoch 22/30
100/100              13s 127ms/step -
accuracy: 0.9580 - loss: 0.0981 - val_accuracy: 0.9660 - val_loss: 0.0858
Epoch 23/30
100/100              13s 133ms/step -
accuracy: 0.9520 - loss: 0.1019 - val_accuracy: 0.9660 - val_loss: 0.0854
Epoch 24/30
100/100              13s 125ms/step -
accuracy: 0.9620 - loss: 0.0964 - val_accuracy: 0.9640 - val_loss: 0.0869
Epoch 25/30
100/100              13s 125ms/step -
accuracy: 0.9635 - loss: 0.0962 - val_accuracy: 0.9660 - val_loss: 0.0852
Epoch 26/30
100/100              22s 138ms/step -
accuracy: 0.9585 - loss: 0.1060 - val_accuracy: 0.9660 - val_loss: 0.0848
Epoch 27/30
100/100              13s 129ms/step -
accuracy: 0.9605 - loss: 0.0913 - val_accuracy: 0.9640 - val_loss: 0.0853
Epoch 28/30
100/100              13s 130ms/step -
accuracy: 0.9680 - loss: 0.0830 - val_accuracy: 0.9660 - val_loss: 0.0839
Epoch 29/30
100/100              12s 123ms/step -
accuracy: 0.9590 - loss: 0.1024 - val_accuracy: 0.9620 - val_loss: 0.0871
Epoch 30/30
100/100              12s 122ms/step -
accuracy: 0.9635 - loss: 0.0818 - val_accuracy: 0.9680 - val_loss: 0.0848
```

```
Restoring model weights from the end of the best epoch: 28.

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or
`keras.saving.save_model(model)`. This file format is considered legacy. We
recommend using instead the native Keras format, e.g.
`model.save('my_model.keras')` or `keras.saving.save_model(model,
'my_model.keras')`.

Q4 Comparison Table

                       model_type  backbone  num_layers  num_params  \
0                   Xception (Q4)  Xception         132    20861480
1         Best VGG16/VGG19 from Q2     VGG19          22    20024384
2  Best augmentation VGG16 from Q1     VGG16          19    14714688

   final_val_acc  final_test_acc  \
0          0.966           0.964
1          0.872           0.910
2          0.898           0.900

                                      model_h5_path
0  /home/solyvie/workspace/cs-projects/25263CSDMa…
1  /home/solyvie/workspace/cs-projects/25263CSDMa…
2  /home/solyvie/workspace/cs-projects/25263CSDMa…

Best overall in Q4 comparison: Xception (test_acc=0.9640)
```



1.5 Example Classification on Unseen/Test Data

[103]:
```
candidate_runs = [
    ('Q2_VGG16', q2_vgg16),
    ('Q2_VGG19', q2_vgg19),
    ('Q4_Xception', q4_xception),
]
# best_name, best_run = sorted(candidate_runs, key=lambda x:␣
↪x[1]['final_test_acc'], reverse=True)[0]
```

```
# best_model = best_run['model_obj']

all_test_images = []
for cls in ['cats', 'dogs']:
    cls_dir = test_dir / cls
    all_test_images.extend([(p, cls) for p in cls_dir.iterdir() if p.is_file()])

img_path, true_label = random.choice(all_test_images)
img = keras_image.load_img(str(img_path), target_size=IMG_SIZE)
img_arr = keras_image.img_to_array(img) / 255.0
img_arr = np.expand_dims(img_arr, axis=0)

img_path, true_label = random.choice(all_test_images)
img = keras_image.load_img(str(img_path), target_size=IMG_SIZE)
img_arr = keras_image.img_to_array(img) / 255.0
img_arr = np.expand_dims(img_arr, axis=0)

pred_lines = []
print('Example image path:', img_path)
print('True label:', true_label)

for model_name, run in candidate_runs:
    model = run['model_obj']
    pred_score = float(model.predict(img_arr, verbose=0)[0][0])  # dog␣
 ↪probability
    pred_label = 'dog' if pred_score >= 0.5 else 'cat'
    confidence = pred_score if pred_label == 'dog' else (1 - pred_score)
    pred_lines.append(f'{model_name}: {pred_label} ({confidence:.3f})')
    print(f'{model_name} -> pred: {pred_label}, dog_prob: {pred_score:.4f},␣
 ↪conf: {confidence:.4f}')

plt.figure(figsize=(6, 5))
plt.imshow(img)
plt.title(
    f"True: {true_label}\n" + "\n".join(pred_lines)
)
plt.axis('off')
plt.show()
```

```
Example image path: /home/solyvie/workspace/cs-
projects/25263CSDMachineLearning/[7] Group 8 - Module 6 CNN Part 3 - Using Pre-
Trained Networks/data/working_split_for_lab3/test/cats/5139.jpg
True label: cats
Q2_VGG16 -> pred: cat, dog_prob: 0.0008, conf: 0.9992
Q2_VGG19 -> pred: cat, dog_prob: 0.0022, conf: 0.9978
Q4_Xception -> pred: cat, dog_prob: 0.0001, conf: 0.9999
```

1.6 Insights

[104]:
```
summary = {
    'q1': q1_df.drop(columns=['history', 'model_obj'], errors='ignore').
 ↪to_dict(orient='records'),
    'q2': q2_df.drop(columns=['history', 'model_obj'], errors='ignore').
 ↪to_dict(orient='records'),
    'q3': q3_df.to_dict(orient='records'),
    'q4': {
        'xception': {k: v for k, v in q4_xception.items() if k not in␣
 ↪['history', 'model_obj']},
        'comparison_table': q4_compare.to_dict(orient='records'),
    },
    'dataset_split_note': split_note,
}

summary_json = OUTPUT_DIR / 'Group8_ANN_Lab3_summary.json'
with open(summary_json, 'w', encoding='utf-8') as f:
```

```
    import json
    json.dump(summary, f, indent=2)

q1_csv = OUTPUT_DIR / 'Group8_ANN_Lab3_q1_table.csv'
q2_csv = OUTPUT_DIR / 'Group8_ANN_Lab3_q2_table.csv'
q3_csv = OUTPUT_DIR / 'Group8_ANN_Lab3_q3_table.csv'
q4_csv = OUTPUT_DIR / 'Group8_ANN_Lab3_q4_table.csv'

q1_df.drop(columns=['history', 'model_obj'], errors='ignore').to_csv(q1_csv,␣
 ↪index=False)
q2_df.drop(columns=['history', 'model_obj'], errors='ignore').to_csv(q2_csv,␣
 ↪index=False)
q3_df.to_csv(q3_csv, index=False)
q4_compare.to_csv(q4_csv, index=False)

q1_best = q1_df.sort_values('final_test_acc', ascending=False).iloc[0]
q2_best = q2_df.sort_values('final_test_acc', ascending=False).iloc[0]
q3_best = q3_df.sort_values('final_test_acc', ascending=False).iloc[0]
q4_best = q4_compare.sort_values('final_test_acc', ascending=False).iloc[0]

q3_sorted = q3_df.sort_values('samples_per_epoch_target').reset_index(drop=True)
q3_sorted['delta_test_acc'] = q3_sorted['final_test_acc'].diff()

p1 = (
    f"For ImageDataGenerator tuning, the stronger result came from␣
 ↪**{q1_best['model_name']}** "
    f"with test accuracy **{q1_best['final_test_acc']:.4f}**, showing that on␣
 ↪this split, its augmentation intensity "
    f"struck a better balance between regularization and preserving␣
 ↪recognizable cat/dog structure. "
    f"Both Q1 runs reached 30 epochs, and the accuracy-loss curves indicate␣
 ↪stable convergence without severe divergence "
    f"between training and validation trajectories."
)

p2 = (
    f"In the VGG comparison, **{q2_best['backbone']}** achieved the higher test␣
 ↪accuracy (**{q2_best['final_test_acc']:.4f}**). "
    f"Architecturally, VGG19 is deeper and has more parameters than VGG16, and␣
 ↪in this run that additional capacity "
    f"translated to slightly better holdout performance. The paired train/
 ↪validation loss and accuracy plots provide the "
    f"same training conditions across both backbones, so the gap is directly␣
 ↪attributable to backbone behavior under the same pipeline."
)
```

```
p3 = (
    f"For the no-augmentation sample-budget experiment, the best observed point␣
↪was **{int(q3_best['samples_per_epoch_target'])} samples/epoch** "
    f"with test accuracy **{q3_best['final_test_acc']:.4f}**. The per-run␣
↪curves and marginal-delta table show non-monotonic gains, "
    f"which suggests diminishing returns and instability when repeatedly␣
↪cycling limited unique images without augmentation. "
    f"Q3 logs were run in silent mode (`verbose=0`) to reduce PDF clutter while␣
↪keeping full metric outputs and plots."
)

p4 = (
    f"In the final architecture comparison, **{q4_best['backbone']}** was best␣
↪overall with test accuracy "
    f"**{q4_best['final_test_acc']:.4f}**. Relative to the best VGG result,␣
↪Xception benefited from depthwise-separable convolution blocks "
    f"and more efficient feature extraction. The final recommendation for this␣
↪notebook is therefore Xception for best predictive performance, "
    f"with Q1/Q2/Q3 retained as ablation evidence and comparative analysis."
)

display(Markdown(p1 + '\n\n' + p2 + '\n\n' + p3 + '\n\n' + p4))

print('Saved summary JSON:', summary_json)
print('Saved tables:')
print(' -', q1_csv)
print(' -', q2_csv)
print(' -', q3_csv)
print(' -', q4_csv)
print('\nQ3 marginal gains table:')
display(q3_sorted[['samples_per_epoch_target', 'final_test_acc',␣
↪'delta_test_acc']])
```

For ImageDataGenerator tuning, the stronger result came from **Q1_VGG16_default_aug** with test accuracy **0.9000**, showing that on this split, its augmentation intensity struck a better balance between regularization and preserving recognizable cat/dog structure. Both Q1 runs reached 30 epochs, and the accuracy-loss curves indicate stable convergence without severe divergence between training and validation trajectories.

In the VGG comparison, **VGG19** achieved the higher test accuracy (**0.9100**). Architecturally, VGG19 is deeper and has more parameters than VGG16, and in this run that additional capacity translated to slightly better holdout performance. The paired train/validation loss and accuracy plots provide the same training conditions across both backbones, so the gap is directly attributable to backbone behavior under the same pipeline.

For the no-augmentation sample-budget experiment, the best observed point was **2000 samples/epoch** with test accuracy **0.9020**. The per-run curves and marginal-delta table show non-monotonic gains, which suggests diminishing returns and instability when repeatedly cycling limited

unique images without augmentation. Q3 logs were run in silent mode (`verbose=0`) to reduce PDF clutter while keeping full metric outputs and plots.

In the final architecture comparison, **Xception** was best overall with test accuracy **0.9640**. Relative to the best VGG result, Xception benefited from depthwise-separable convolution blocks and more efficient feature extraction. The final recommendation for this notebook is therefore Xception for best predictive performance, with Q1/Q2/Q3 retained as ablation evidence and comparative analysis.

```
Saved summary JSON: /home/solyvie/workspace/cs-
projects/25263CSDMachineLearning/[7] Group 8 - Module 6 CNN Part 3 - Using Pre-
Trained Networks/models/Group8_ANN_Lab3_summary.json
Saved tables:
 - /home/solyvie/workspace/cs-projects/25263CSDMachineLearning/[7] Group 8 -
Module 6 CNN Part 3 - Using Pre-Trained
Networks/models/Group8_ANN_Lab3_q1_table.csv
 - /home/solyvie/workspace/cs-projects/25263CSDMachineLearning/[7] Group 8 -
Module 6 CNN Part 3 - Using Pre-Trained
Networks/models/Group8_ANN_Lab3_q2_table.csv
 - /home/solyvie/workspace/cs-projects/25263CSDMachineLearning/[7] Group 8 -
Module 6 CNN Part 3 - Using Pre-Trained
Networks/models/Group8_ANN_Lab3_q3_table.csv
 - /home/solyvie/workspace/cs-projects/25263CSDMachineLearning/[7] Group 8 -
Module 6 CNN Part 3 - Using Pre-Trained
Networks/models/Group8_ANN_Lab3_q4_table.csv

Q3 marginal gains table:
```

	samples_per_epoch_target	final_test_acc	delta_test_acc
0	500	0.888	NaN
1	1000	0.896	0.008
2	2000	0.902	0.006
3	5000	0.898	-0.004
4	10000	0.900	0.002
5	15000	0.890	-0.010